

$$p = \frac{C_L V^2}{1,000} \quad (3.7.3.2-1)$$

where:

p = lateral pressure (ksf)

C_L = lateral drag coefficient, specified in Table 3.7.3.2-1

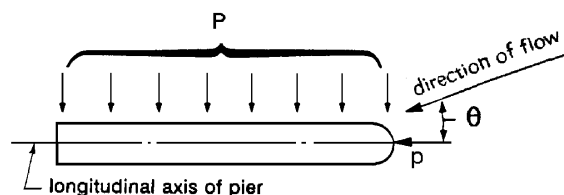


Figure 3.7.3.2-1—Plan View of Pier Showing Stream Flow Pressure

Table 3.7.3.2-1—Lateral Drag Coefficient

Angle, θ , between direction of flow and longitudinal axis of the pier	C_L
0 degrees	0.0
5 degrees	0.5
10 degrees	0.7
20 degrees	0.9
≥ 30 degrees	1.0

The lateral drag force shall be taken as the product of the lateral stream pressure and the surface exposed thereto.

3.7.4—Wave Load

Wave action on bridge structures shall be considered for exposed structures where the development of significant wave forces may occur.

C3.7.4

Loads due to wave action on bridge structures shall be determined using accepted engineering practice methods. Site-specific conditions should be considered. The latest editions of *AASHTO Guide Specifications for Bridges Vulnerable to Coastal Storms*, the *Coastal Engineering Manual*, published by the U.S. Army Engineer Research and Development Center, or FHWA's *Hydraulic Engineering Circular 25* (HEC-25) may be utilized for the computation of wave forces.

3.8—WIND LOAD: WL AND WS

3.8.1—Horizontal Wind Load

3.8.1.1—Exposure Conditions

3.8.1.1.1—General

Wind pressure shall be assumed to be uniformly distributed on the area exposed to the wind. The exposed area shall be the sum of areas of all components, including floor system, railing, and sound barriers, as seen in elevation taken perpendicular to the wind direction. The wind load shall be the product of

C3.8.1.1.1

For most bridges, the same wind pressure will be used for all components of the structure and the wind load is applied as a uniformly distributed load on the entire exposed area of the structure. However, some situations may warrant using different wind pressures on different components. The most common cases are for